PHY202: Physics II: Exam 7
Dr. Deskins

$\frac{12}{12} \Rightarrow 100\%$

0	4	8	13	17	21	25	29	33	38	42	46	50	54	58	63	67	71	75	79	83	88	92	96	100
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12

Name:

These questions assess your understanding of the material from Chapters 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Visible light of wavelength 430 nm falls on a single slit and produces its second diffraction minimum at an angle of 33.9 relative to the incident direction of the light. What is the width of the slit?

ANSWER: $1.6 \cdot 10^{-6}$ m

$d\sin\theta = m\lambda$

$d = \frac{m\lambda}{\sin\theta} = \frac{2 \cdot 4.3 \cdot 10^{-7} m}{\sin 33°} = 1.6 \cdot 10^{-6} m$

$\lambda = 4.30 \cdot 10^{-7} m$
$\theta = 33°$
$m = 2$

✓

2. Suppose a space probe is moving away from Earth at a speed of 1.1×10^8 m/s. It sends a radio-wave message back to Earth at a frequency of 4.8 GHz. At what frequency is the message received on Earth?

ANSWER: $3.3 \cdot 10^9$ Hz

$f_{obs} = f_s \sqrt{\frac{1 - \frac{u}{c}}{1 + \frac{u}{c}}}$

$f_{obs} = 4.8 \cdot 10^9 Hz \sqrt{\frac{1 - 0.366}{1 + 0.366}}$

$f_{obs} = 3.3 \cdot 10^9 Hz$

$\frac{u}{c} = \frac{1.1 \cdot 10^8 \frac{m}{s}}{3.0 \cdot 10^8 \frac{m}{s}} = 0.366$

$f_s = 4.8 \cdot 10^9 Hz$

✓

3. You shine a beam of white light through a diffraction grating with 8.7×10^3 lines per centimeter to a screen 2.0 m away. Find the angle for first-order diffraction of 550 nm light.

ANSWER: $29°$

$\frac{8.7 \cdot 10^3 \text{ lines}}{0.01 \text{ m}} = \frac{8.7 \cdot 10^5 \text{ lines}}{1 \text{ m}} \quad \frac{1 \text{ m}}{8.7 \cdot 10^5 \text{ lines}} = 1.149 \cdot 10^{-6} m$

$m = 1$
$\lambda = 5.5 \cdot 10^{-7} m$

$\theta = \sin^{-1}\left(\frac{m\lambda}{d}\right)$

$\theta = \sin^{-1}\left(\frac{1 \cdot 5.5 \cdot 10^{-7} m}{1.149 \cdot 10^{-6} m}\right) = 29°$

✓

+3

4. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be 430 nm, $n_{\text{oil}} = 1.44$ and $n_{\text{water}} = 1.32$.

$2nt = (m + \frac{1}{2})\lambda$

$t = \frac{\lambda}{4n} = \frac{4.30 \cdot 10^{-7} m}{4 \cdot 1.44} = 7.5 \cdot 10^{-8} m$

ANSWER: $7.5 \cdot 10^{-8}$ m

$m = 0$

$\lambda = 4.3 \cdot 10^{-7} m$

$n = 1.44$

5. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 1.2 kW/m² to reduce the intensity to 0.49 kW/m²?

$I = I_0 \cos^2\theta$

$\cos^{-1}\sqrt{\frac{I}{I_0}} = \theta$

$\cos^{-1}\sqrt{\frac{0.49 \, kW/m^2}{1.2 \, kW/m^2}} = 50.°$

ANSWER: 50.°

$I = 0.49 \, kW/m^2$

$I_0 = 1.2 \, kW/m^2$

6. Suppose a galaxy is moving away from Earth at a speed of 1.9×10^8 m/s. It emits radio waves with a wavelength of 0.84 m. What wavelength would we detect on Earth?

$\lambda_{obs} = \lambda_s \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}$

$\lambda_{obs} = 0.84 m \sqrt{\frac{1 + 0.633}{1 - 0.633}}$

$\lambda_{obs} = 1.8 m$

ANSWER: 1.8 m

$\frac{u}{c} = \frac{1.9 \cdot 10^8 \frac{m}{s}}{3.0 \cdot 10^8 \frac{m}{s}} = 0.633$

$\lambda_s = 0.84 m$

7. Suppose you pass light from a laser through two slits separated by 6.4×10^{-3} mm and find that the third bright line on a screen is formed at an angle of 8.6° relative to the incident beam. What is the wavelength of the light?

$\lambda = \frac{d \sin\theta}{m}$

$\lambda = \frac{6.4 \cdot 10^{-6} m \cdot \sin 8.6°}{3}$

$\lambda = 3.2 \cdot 10^{-7} m$

ANSWER: $3.2 \cdot 10^{-7}$ m

$d = 6.4 \cdot 10^{-6} m$

$m = 3$

$\theta = 8.6°$

+4

8. The primary mirror of an orbiting space telescope has a diameter of 1.3 m. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of 530 nm, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.

$\theta = 1.22 \frac{\lambda}{D} = 1.22 \frac{5.3 \cdot 10^{-7} m}{1.3 m} = 4.974 \cdot 10^{-7}$ rad

ANSWER: 0.10 arcseconds

$4.974 \cdot 10^{-7} rad \cdot \frac{180°}{\pi \, rad} \cdot \frac{60 \, arcmin}{1°} \cdot \frac{60 \, arcsec}{1 \, arcmin} =$

0.10 arcseconds

$\lambda = 5.3 \cdot 10^{-7} m$

$D = 1.3 m$

✓

9. The headlights of a car have a wavelength of about 640 nm and are 1.5-m apart. Assuming the eye's pupil diameter is 0.75 cm, what is the maximum distance at which the two headlights can be resolved? Use the diffraction limit (Rayleigh criterion) for your calculation.

$\theta = 1.22 \frac{\lambda}{D} = 1.22 \frac{6.4 \cdot 10^{-7} m}{7.5 \cdot 10^{-3} m} = 1.041 \cdot 10^{-4}$ rad

ANSWER: $1.4 \cdot 10^{4}$ m

L = max Length

$\theta = \frac{s}{L}$

$L = \frac{s}{\theta} = \frac{1.5 m}{1.041 \cdot 10^{-4} rad} = 1.4 \cdot 10^{4} m$

$D = 7.5 \cdot 10^{-3} m$

$\lambda = 6.4 \cdot 10^{-7} m$

$s = 1.5 m$

✓

10. Calculate the energy that is released upon the complete annihilation of a 13.-g object.

ANSWER: $1.2 \cdot 10^{15}$ J

$E_0 = mc^2$

$E_0 = 0.013 kg \cdot (3.0 \cdot 10^{8} \frac{m}{s})^2$

$E_0 = 1.2 \cdot 10^{15} J$

$m = 0.013 kg$

✓

11. Suppose an astronaut is traveling at 2.0×10^8 m/s from Earth to Alpha Centauri, which is a star system 2.1 ly away as measured by an Earth-bound observer. How far apart are Earth and Alpha Centauri as measured by the astronaut? Note: ly is lightyear, the distance light travels in one year.

$L = L_0 \left(\sqrt{1 - \frac{v^2}{c^2}}\right)$

ANSWER: 1.6 ly

$\frac{v}{c} = \frac{2.0 \cdot 10^{8} \frac{m}{s}}{3.0 \cdot 10^{8} \frac{m}{s}} = 0.66$

$L = 2.1 ly \left(\sqrt{1 - (0.66)^2}\right)$

$L = 1.6 ly$

$L_0 = 2.11 ly$

✓

+4

12. Suppose a cosmic ray colliding with a nucleus in Earth's upper atmosphere produces a muon that has a velocity equal to 1.0×10^8 m/s. The muon then travels at constant velocity and lives for a duration of $1.0\,\mu s$ in the muon's frame of reference. How long does the muon live as measured by an Earth-bound observer?

ANSWER: $1.1 \cdot 10^{-6}$ s

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{1}{\sqrt{1-(0.33)^2}} = 1.0607$$

$\Delta t_0 = 1.0 \cdot 10^{-6}$ s

$$\frac{v}{c} = \frac{1.0 \cdot 10^8 \frac{m}{s}}{3.0 \cdot 10^8 \frac{m}{s}} = 0.\overline{33}$$

$\Delta t = \gamma \Delta t_0$

$\Delta t = 1.0607 \cdot (1.0 \cdot 10^{-6} s)$

$\Delta t = 1.1 \cdot 10^{-6}$ s

+1

PHY202: Physics II: Exam 7

1. $D = 1.58 \times 10^{-6}$ m $\because D \sin\theta = m\lambda$
2. $f = 3.27 \times 10^{9}$ Hz $\because \lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f\lambda$
3. $\theta = 28.6°$ $\because d \sin\theta = m\lambda$
4. $t = 7.47 \times 10^{-8}$ m $\because 2t = \left(m + \frac{1}{2}\right)\lambda_n$, $n_{\text{air}} < n_{\text{oil}} > n_{\text{water}}$
5. $\theta = 50.3°$ $\because I = I_0 \cos^2\theta$
6. $\lambda_{obs} = 1.77\,\text{m}$ $\because \lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$
7. $\lambda = 3.19 \times 10^{-7}$ m $\because d \sin\theta = m\lambda$
8. $\theta = 0.103$ arcseconds $\because \theta = 1.22\frac{\lambda}{D}$
9. $x = 1.44 \times 10^{4}$ m $\because \theta = 1.22\frac{\lambda}{D}$, $\Delta\theta = \frac{\Delta s}{r}$, $\Delta s \approx 1.5\,\text{m}$ for small angles
10. $E = 1.17 \times 10^{15}$ J $\because E_0 = mc^2$
11. $L = 1.57\,\text{ly}$ $\because L = L_0 \gamma^{-1}$
12. $\Delta t = 1.06 \times 10^{-6}$ s $\because \gamma = \left(\sqrt{1 - \frac{v^2}{c^2}}\right)^{-1}$, $\Delta t = \gamma \Delta t_0$